MiniShop v1 — Test Plan

Application	MiniShop v1: single-file web app `mini-shop.html` (no backend, all state in page memory)
Basis	Functional Requirements R1–R11 in the assignment brief
Executed	2026-09-27, macOS, Chromium 153 over `file://` (both failing cases also reproduced in Firefox and WebKit)
Result	23 cases executed: **18 Pass · 2 Fail · 3 undefined by spec**. 2 bugs logged, 10 spec gaps raised

1. Scope

In scope

- Every functional requirement R1–R11: login, lockout, product list, cart badge, empty and non-empty checkout, confirmation, logout and Back to Shop.
- Negative and edge behaviour around those requirements: lockout boundaries, what counts as "consecutive", empty and whitespace input, case sensitivity, repeat adds, high-volume totals, and state carried across logout.
- Light exploratory checks (keyboard, accessibility basics, client-side security, refresh behaviour). Anything found here is recorded as an **observation**, not a requirement bug, because v1 has no non-functional requirements.

Out of scope

- The backend and API. There is none for the UI; the API contract is covered in Part B.
- Performance and load, localisation, mobile and responsive layouts, and formal WCAG audit.
- Browsers other than Chromium, Firefox and WebKit (the Playwright engines).

2. Approach

Technique	Where it is used
Requirement-based testing	Each requirement maps to one or more cases (see section 6, traceability)
Boundary value analysis	Lockout at 2 / 3 / 4 attempts (TC-08, TC-09)
State transition testing	Screen flow and lockout state across login, logout and checkout (state-transition table below; TC-10, TC-19, TC-22)
Equivalence partitioning	Wrong password, unknown user, empty fields, case and whitespace variants (TC-03 to TC-07)
Error guessing and exploratory testing	Non-consecutive failures, failures on other usernames, page reload, 24-item cart, stale messages (TC-10 to TC-12, TC-20, TC-23, section 5)

State transitions (screens and lockout state):

From	Transition (event, result)	To
Start	Page opened	Login
Login	Valid credentials (not locked)	Shop
Login	Invalid (failures < 3), "Invalid username or password."	Login
Login	3rd consecutive failure	Locked
Locked	Any attempt, "Account locked..."	Locked
Shop	Checkout (cart not empty), cart reset to 0	Confirmation
Shop	Checkout (cart empty), alert "Your cart is empty."	Shop

From	Transition (event, result)	To
Confirmation	Back to Shop	Shop
Shop	Logout (fields cleared)	Login

Execution. Cases were run in order, each from a fresh page load, in Chromium 153 on macOS. For every case, the exact on-screen text, alert text and element visibility were recorded as the actual result. The automatable cases also live in the Part D regression suite (`automation/`), which runs on Chromium, Firefox and WebKit.

Entry criteria: the app opens from `file://` with no console errors, and the requirements are baselined as v1. **Exit criteria:** every case is executed, every failure is logged with repro steps, severity and priority, and spec gaps are handed to the Product Owner.

3. Assumptions

1. The app is used exactly as supplied. The only edit was rejoining lines the PDF had wrapped (for example `Add to` / `Cart`), which changes no behaviour.
2. **"Consecutive"** (R5) means failed attempts with no successful login in between. This is the standard meaning, so a successful login resets the count.
3. Every case starts from a **fresh page load**. All state (cart, failure counter, lock) lives in page memory and is wiped by a reload.
4. Where the requirements are silent, the case records what the app actually does and marks the result **"Undefined by spec"** instead of inventing an expected result. These cases feed the spec gaps in section 7.
5. Severity and priority follow the definitions in `bug-reports.xlsx` (Legend sheet).

4. Test case list

The full test cases (preconditions, test data, steps, expected and actual results, status and notes) are in **`manual-test-cases.pdf`** in this folder. The list below is the index.

Status key: **Pass** · **Fail** (bug logged) · **Undefined** (the spec is silent; actual behaviour recorded and a spec gap raised).

ID	Title	Req	Type	Pri	Status
Login (R1–R4)					
TC-01	Login screen is shown by default	R1	Positive	P1	**Pass**
TC-02	Valid credentials open the shop	R2, R3	Positive	P1	**Pass**
TC-03	Wrong password is rejected	R4	Negative	P1	**Pass**
TC-04	Unknown username is rejected	R4	Negative	P2	**Pass**
TC-05	Empty username and password	R4, R5	Negative	P2	**Pass** (raises SG-04)
TC-06	Credentials are case-sensitive	R2	Edge	P3	**Pass** (raises SG-03)
TC-07	Leading and trailing whitespace is not trimmed	R2	Edge	P3	**Pass** (raises SG-03)
Lockout (R4, R5)					
TC-08	Lockout boundary: messages on attempts 1–3	R4, R5	Boundary	P1	**Fail** (**BUG-A02** (requirement conflict, SG-01))
TC-09	Correct credentials are refused while locked	R5	Negative	P1	**Pass**
TC-10	Non-consecutive failures do not lock the account	R5	State / Edge	P1	**Fail** (**BUG-A01**)
TC-11	Failures on *other* usernames and the valid account	R5	Edge	P2	**Undefined** (SG-02)
TC-12	Lockout survives a page reload	R5	Edge	P2	**Undefined** (SG-02)
Product list and cart (R6, R7)					

ID	Title	Req	Type	Pri	Status
TC-13	Shop lists 3 products	R6	Positive	P1	**Pass**
TC-14	Badge starts at 0 and +1 per click	R7	Positive	P1	**Pass**
TC-15	Same item added multiple times	R7, R9	Edge	P2	**Pass** (raises SG-06)
Checkout and confirmation (R8, R9, R11)					
TC-16	Checkout with an empty cart	R8	Negative	P1	**Pass**
TC-17	Checkout with a single item	R9	Positive	P1	**Pass**
TC-18	Checkout with one of each product	R9	Positive	P1	**Pass**
TC-19	Back to Shop, and the cart really is emptied	R8, R9, R11	State	P1	**Pass**
TC-20	High-volume cart total accuracy	R7, R9	Edge	P3	**Pass** (see OBS-06)
Logout and session (R10)					
TC-21	Logout returns to login and clears fields	R10	Positive	P1	**Pass**
TC-22	Cart after logout and re-login	R10	Edge	P2	**Undefined** (SG-05)
TC-23	No stale error message after logout	R4, R10	Edge	P3	**Pass**

5. Exploratory session notes

A time-boxed charter: "*Explore login and session handling with keyboard, reload and DevTools to find ways the stated behaviour can be bypassed or is hard to use.*"

Finding	Notes	Logged as
Valid credentials sit in the page source	`VALID_USER` / `VALID_PASS` are plain JavaScript. View Source reveals `secret123`	OBS-01
Lockout is client-side only	Reload (TC-12), or setting `locked = false` / `failedAttempts = 0` in the console, bypasses it. Unlimited brute force is possible	OBS-02
Enter doesn't submit the login	There is no `<form>`, so the user must click Login	OBS-03
No labels or live region	Inputs rely on placeholder text only, and `#error` isn't an `aria-live` region, so screen-reader users don't hear login errors	OBS-04
Reload while shopping	Reloading the shop logs the user out and silently empties the cart	OBS-05 (SG-09)
Floating-point money	Totals are summed as floats and rounded with `toFixed(2)`. Correct for every tested combination, but fragile	OBS-06

Observations are in the **Observations** sheet of `bug-reports.xlsx`. They are real quality risks, but they break no stated v1 requirement, so they are deliberately **not** filed as requirement bugs.

6. Requirement traceability

Req	Summary	Test cases	Result
R1	Login screen by default	TC-01	Pass
R2	Valid credentials	TC-02, TC-06, TC-07	Pass
R3	Login shows shop	TC-02	Pass
R4	Invalid-login message	TC-03, TC-04, TC-05, TC-08, TC-23	**TC-08 fails (BUG-A02)**
R5	Lock after 3 consecutive failures	TC-05, TC-08, TC-09, TC-10, TC-11, TC-12	**TC-10 fails (BUG-A01)**. TC-11 and TC-12 undefined

Req	Summary	Test cases	Result
R6	3 products listed	TC-13	Pass
R7	Add to Cart increments badge	TC-14, TC-15, TC-20	Pass
R8	Empty-cart alert	TC-16, TC-19	Pass
R9	Confirmation count and total, badge reset	TC-15, TC-17, TC-18, TC-19, TC-20	Pass
R10	Logout returns to login and clears fields	TC-21, TC-22, TC-23	Pass. TC-22 undefined
R11	Back to Shop	TC-19	Pass

Execution summary: 23 executed. 18 Pass (TC-05, TC-06, TC-07 and TC-15 pass but surface spec gaps), 2 Fail (TC-08 → BUG-A02, TC-10 → BUG-A01), 3 Undefined by spec (TC-11, TC-12, TC-22).

7. Requirement gaps and ambiguities

These are problems with the **requirements**, not the code. Where the code's behaviour is merely one reasonable reading of a silent spec, it isn't called a bug. The PO needs to decide the expected behaviour, and the cases above get updated to match.

ID	Req	Gap or ambiguity	Current app behaviour	Why it matters / question for the PO
SG-01	R4 vs R5	**Conflict.** R4 says *every* invalid attempt shows "Invalid username or password." R5 says the lock message appears on *further* attempts. The 3rd failure is both an invalid attempt and the locking attempt	3rd failure shows the lock message	Which message does the 3rd attempt show? Showing the lock immediately is arguably better UX. If so, reword R5 to "the 3rd consecutive failure locks the account and shows…". Tracked as BUG-A02 until decided
SG-02	R5	**Lock scope, duration and unlock path are undefined.** Whose account locks when failures use other usernames? How long does the lock last? How is it lifted? Should it survive a reload?	Global per-tab lock (TC-11). Permanent until reload (TC-12). No unlock path	Without these answers, R5 can't be tested for correctness. Also a security question (OBS-02)
SG-03	R2	Matching rules for credentials: case sensitivity (especially the username), whitespace trimming, length and character limits	Exact, case-sensitive match. No trimming	Usernames are usually case-insensitive and trimmed. Pasting a trailing space currently burns a lockout attempt
SG-04	R4, R5	**Empty-field handling.** Should submitting empty fields show field-level validation ("Username is required") and *not* count toward the lock?	Counts as a failed attempt. 3 empty clicks lock the account	An accidental triple-click locks the only account
SG-05	R10	**Does logout end the session's cart?** R10 only mentions the fields	Cart and badge survive logout and re-login (TC-22)	On a shared machine, the next person inherits and can buy the previous user's cart
SG-06	R7	Cart model is undefined: does the badge count units or distinct products? Is there a maximum quantity? Can items be viewed, decremented or removed?	Badge counts units. Each click adds a separate line. No cart view or removal	A user who over-adds can't correct it without checking out or reloading
SG-07	R9	Confirmation content is under-specified: count = units or lines, pluralisation, currency, tax and shipping, order number, itemised list	"N item(s), total $X.XX". No item list or order number	"1 item(s)" reads poorly, and the user can't verify *what* was ordered

ID	Req	Gap or ambiguity	Current app behaviour	Why it matters / question for the PO
SG-08	R6	**Stock is missing from the UI spec.** The Part B API marks USB-C Hub `inStock: false` and rejects orders for it with 409, but R6 and R7 have no out-of-stock state	USB-C Hub can be added and bought	Once wired to the API, users would hit a 409 at checkout. The UI needs a disabled or out-of-stock state
SG-09	—	Session persistence isn't specified: reload, browser back, a new tab, and whether the confirmation screen should offer Logout	Reload returns to login and loses the cart and lock. The confirmation screen has no Logout	Needed to define expected behaviour for those cases
SG-10	—	**No non-functional requirements:** accessibility (labels, keyboard submit, announced errors), supported browsers, security (where credentials are checked and stored)	See OBS-01 to OBS-04	Without NFRs, the observations can't be raised as requirement bugs, though they would block a real release